# META RADAR

**From Scattered Signals to Strategic Intelligence**

**Pilot Area:** Haemophilia A & B (Rare Disease)
**Novo Nordisk GBS Hackathon 2026** — Problem Statement #3


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## 1. Problem & Relevance to Novo Nordisk

The haemophilia treatment landscape is evolving rapidly across replacement factors, non-factor therapies (*emicizumab*, *concizumab*, investigational *denecimig [Mim8]*), RNAi approaches (*fitusiran*), tissue factor pathway inhibitors (*marstacimab*), and gene therapies such as *Hemgenix*, alongside the evolving gene-therapy landscape. Critical external signals are fragmented across scientific literature, clinical trial registries, press releases, regulatory announcements, safety advisories, and medical congress readouts. Cross-functional teams spend hours scouring scattered feeds and receiving generic document summaries rather than connected development insights.

**MetaRadar** solves this challenge by converting raw external signals into source-linked, prioritized, role-relevant intelligence. It is **NOT a standard news dashboard**. Instead of broadcasting every headline to everyone, MetaRadar frames intelligence around four decision-first questions:

**(1) What changed?** → **(2) Why does it matter?** → **(3) Who should review it?** → **(4) What action may be needed?**

**Intended Users:** Cross-functional Novo Nordisk teams spanning **Medical Affairs, Regulatory, Safety / Pharmacovigilance, Market Access, and Leadership**. MetaRadar is designed strictly to support human decision-making and accelerate expert review, not to replace professional judgment.

## 2. Overall Concept & Core Capabilities

MetaRadar is an AI-enabled competitive intelligence radar that ingests scattered public updates, connects related evidence into unified timelines, and routes actionable insights to specific functions. Key capabilities include:

- **Signal Detection & Classification:** Ingests external updates and classifies them across signal types (clinical trials, publications, news, regulatory, safety, congresses, market/access) and haemophilia entities (disease type, factor, inhibitor status, company, drug modality).
- **Evidence Linking & Summarization:** Aggregates related signals across sources into connected evidence chains and generates source-grounded summaries with explicit **FACT** / **INTERPRETATION** / **SPECULATION** labeling.
- **Prioritization & Functional Routing:** Computes role-specific relevance scores to direct insights to relevant functions (*"not every signal goes to everyone"*) while keeping high-impact cards visible to Leadership.
- **Lifecycle Awareness & Watch-for-Next:** Tracks asset progressions (*Trial* → *Congress Readout* → *Publication* → *Regulatory Filing* → *Access Event*) and flags expected upcoming milestones for human review.
- **Stakeholder Calibration (HITL):** Uses structured expert feedback to calibrate relevance scoring, priority assignment, and functional routing.

## 3. Haemophilia Scope & Data Boundary

- **Primary Focus:** Haemophilia A and Haemophilia B within Rare Disease.
- **Data Boundary: Public data, mock data, and synthetic data ONLY**. Zero confidential Novo Nordisk information and zero patient-identifiable private health information (PII/PHI) are processed or used.
- **Signal Categories:** Clinical trial updates, scientific publications, company press releases, regulatory decisions, safety advisories, congress presentations (ASH, ISTH, WFH, EHA), and market/access updates.
- **Live vs. Planned/Simulated Distinctions:** The prototype integrates live public connectors for PubMed (NCBI), ClinicalTrials.gov, and NewsAPI. Extended categories (EMA RSS feeds, OpenFDA, medical congresses, market access) are configured via adapter-ready or synthetic demo feeds for hackathon demonstration.

## 4. Out of Scope

To maintain clear hackathon execution boundaries, the MetaRadar prototype strictly does **NOT** attempt to:

- Make autonomous clinical, medical, or regulatory decisions, or provide patient-specific medical advice.
- Replace healthcare professionals, medical affairs directors, regulatory leaders, or safety specialists.
- Access, store, or process any confidential Novo Nordisk proprietary data or private patient health data (PII/PHI).
- Automatically execute external business actions or communications without explicit human expert review.
- Serve as a production-grade enterprise deployment during the hackathon phase.

## 5. Data Sources

MetaRadar synthesizes data across three operational source tiers:

- **Prototype-Connected Live Sources:** NCBI PubMed E-utilities (scientific literature), ClinicalTrials.gov APIv2 (trial updates), and NewsAPI (pharma news & announcements).
- **Adapter-Ready Public Sources:** OpenFDA API (regulatory drug approvals) and EMA RSS feeds (European regulatory decisions).
- **Simulated / Synthetic Demo Feeds:** Pre-curated, labelled haemophilia signal datasets for medical congress abstracts, safety advisories, and market access events to guarantee robust offline demonstration.

## 6. Intelligence Pipeline

MetaRadar processes information through a decision-centric workflow:

**Source Ingestion** → **Signal Detection** → **Classification** → **Deduplication** → **Evidence Summarization**
→ **Priority Assessment** → **Functional Routing** → **Human Review** → **Stakeholder Calibration**

Raw payloads are deduplicated via stable source keys, scrubbed for privacy, and normalized against a haemophilia domain ontology. High-level orchestrators link incoming signals into evidence chains, assess triage priority, and generate function-routed decision cards for expert review.

## 7. Source Traceability & Trust

Trust is paramount in pharma competitive intelligence. MetaRadar enforces strict auditability:

- **Source-Linked Evidence:** Every intelligence card maintains direct links to the underlying evidence supporting its factual claims and context (PMID, NCT ID, press release URL, regulatory filing ID).
- **Decision-Support Hypotheses:** Suggested next steps are clearly identified as AI-generated decision-support hypotheses requiring human review, rather than prescriptive corporate actions.
- **Fact vs. Interpretation vs. Speculation:** Intelligence outputs explicitly categorize statements as **FACT** (verifiable source excerpt), **INTERPRETATION** (AI-reasoned contextual implication), or **SPECULATION** (forward-looking hypothesis).
- **Human-in-the-Loop Supervision:** Sensitive medical, regulatory, and safety insights explicitly require human expert review before internal action is taken.

## 8. Stakeholder Calibration & Learning Loop

MetaRadar incorporates a structured Human-in-the-Loop (HITL) feedback mechanism to align system prioritization and routing with organizational preferences over time:

**AI Baseline Output** → **Stakeholder Expert Review** → **Structured Feedback Rating** → **Calibrated Routing Weights** → **Improved Future Outputs**

### 8.1 Concise Before vs. After Calibration Example (Public/Mock Data)

| | |
|---|---|
| **AI Baseline Output** | "Clinical trial activity increased around a competing haemophilia therapy (Phase III readout)." |
| **Calibrated Output** | **Priority: High \| Medical Relevance: High**; monitor trial phase transition, bleed rate endpoint comparisons, and safety updates; route to Medical Affairs and Competitive Intelligence for strategic review. |

Stakeholder calibration adjusts scoring weights, routing rules, and action prompts based on expert ratings; it refines priority and relevance rather than granting autonomous execution authority to the AI.

## 9. User Interface & Decision Panels

The MetaRadar user interface focuses on decision-first clarity through key prototype screens:

- **Radar Overview Dashboard:** Displays active competitive signals, high-confluence clusters, and strategic updates across Haemophilia A & B.
- **Interactive Signal Cards:** Present high-priority developments clearly answering: (1) What changed?, (2) Why it matters?, (3) Triage Priority (High/Medium/Low), (4) Suggested function review, (5) Suggested next step for expert review, and (6) Traceable source evidence links with confidence strength.
- **Search, Filters & Role-Oriented Views:** Allows filtering by drug modality (bispecific, gene therapy, factor replacement), company, source, priority, and functional views tailored for **Medical Affairs, Regulatory, Safety / PV, Market Access, and Leadership**.

## 10. Four-Week Development Timeline (Hackathon Scope)

MetaRadar follows a disciplined, realistic 4-week prototype timeline tailored for a 1-month hackathon:

| Week | Key Deliverables & Milestones |
|---|---|
| **WEEK 1**<br>**Problem & Data Setup** | Haemophilia A/B signal taxonomy; public/mock/synthetic data setup; source connector configuration; core architecture and prototype scope locking. |
| **WEEK 2**<br>**Detection & AI Logic** | Ingestion from public sources; signal detection & classification; evidence linking; deduplication & provenance tracking. |
| **WEEK 3**<br>**Prioritization & UI** | Evidence-linked summarization; priority scoring; functional role routing; stakeholder calibration loop; dashboard & interactive signal cards. |
| **WEEK 4**<br>**Validation & Demo** | End-to-end validation, stakeholder calibration, usability refinement, and final demonstration. |

## 11. Novo Nordisk Input Needed

To ensure maximum strategic alignment during the hackathon, we welcome stakeholder guidance from Novo Nordisk on the following key questions:

1. Which external haemophilia signal categories (e.g., Phase III trial readouts vs. real-world access updates) should receive the highest baseline weighting?
2. How should signal prioritization and routing preferences differ between Medical Affairs, Regulatory, Safety, Market Access, and Leadership?
3. What specific clinical or market criteria should most heavily influence signal urgency scoring?
4. How can stakeholder feedback (HITL) be structured to be most intuitive and non-intrusive for busy pharma professionals?
5. Which dashboard views and decision workflows would provide the highest immediate decision-making value for Novo Nordisk teams?

*Note: These input requests represent valuable alignment guidance to maximize utility, not blocking dependencies for building the functional prototype.*